LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration of Soldering and de-soldering techniques adopted on PCBs and cables.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration of Soldering and de-soldering techniques adopted on PCBs and cables. Safety precautions to be observed while Soldering and De-soldering on PCBs and cables.
5. Specific Objectives	24.13.1.1 Trainee will be able to carry out workshop maintenance on aircraft cables and PCBs.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of soldering and de-soldering on PCBs and cables; Safety precautions; Instructor supervision
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary safety hazards associated with hot soldering irons?
(b) Identify the components required for a basic soldering setup.
(c) Why is flux necessary during the soldering process?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Soldering iron selection and temperature settings • Proper flux application techniques • Heating and melting solder on PCBs • Cooling and inspection of solder joints	L & Ds	PPT, IB	15 min
(b)	• De-soldering techniques using desoldering pump • Heat sink application for component removal • Cleaning flux residue from PCBs • Inspection criteria for damaged traces	L & Ds	PPT, PDF	20 min
(c)	• Safety precautions for fume extraction • Fire prevention measures in workshop • Proper disposal of solder waste • Personal protective equipment usage	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct sequence for applying flux before heating the joint.
(b) Describe the immediate action required if a soldering iron tip oxidizes during use.

12. Home Assignment and Follow Up	Prepare a brief report on the specific safety precautions required when de-soldering high-density aircraft cables, referencing the next lesson on crimping techniques.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration of crimping techniques adopted on aircraft cables. Safety precautions to be observed while carrying out crimping on aircraft cables.	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min